

Duality of Standard Form LP

$$\min: c^T x$$

$$\text{st: } Ax = b$$

$$x \geq 0$$

// using $\begin{matrix} \min & \max \\ \max & \min \end{matrix}$

$\geq$

$$\max: p^T b$$

$$\text{st: } p^T A \leq c$$

p free

Weak Duality:

$\forall$ feasible x

$\forall$ feasible p

$$c^T x \geq p^T b$$

$$\begin{array}{ll} \max: & p^T b \\ \text{st:} & p^T A \leq c^T \\ & p \text{ free.} \end{array}$$

$$\text{BFS: } \hat{x} = \begin{pmatrix} x_B \\ 0 \end{pmatrix}, \quad x_B = A_B^{-1} b, \quad B \subseteq [n], \quad |B| = m.$$

A hand-drawn diagram of a table. The top row is the header, with the text "Reduced Costs" circled in the right cell. The table has two columns and two rows. To the left of the table, there is a circled label $x_B = A_B^{-1}b$ with an arrow pointing to the left cell of the first row. Below the table, there is a small symbol that looks like ΔT .

$$\bar{C} = C - (\overset{\vec{P}}{C_B^T A_B^{-1}}) A$$

$$\forall i \in B, \quad \bar{c}_i = 0.$$

BFS π is optimal

ii) $\bar{c} \geq 0$. Let $\hat{p}^2 = c_B' A_B^{-1}$

$$\hat{x} \text{ is optimal} \Leftrightarrow \hat{p}^T A \leq c$$

$$\Leftrightarrow \hat{p} \text{ is dual feasible.}$$

$$\hat{p}^T b = c_B^T \underbrace{A_B^{-1} b}_{\hat{x}_B} = c_B^T \hat{x}_B = c^T \hat{x}$$

since $\forall p$ feasible, x feasible $p^T b \leq c^T x \Rightarrow \hat{p}, \hat{x}$
 opt

Summary: Strong Duality for LP

(A) If an LP has an opt soln,
then so does its dual.
AND opt. values are equal.

(B) Primal unbbd $\Rightarrow$ Dual infeasible

(D) Comp. slackness.

(C) Simplex alg. reveals primal & dual
opt. soln:

$$x = \begin{pmatrix} x_B \\ 0 \end{pmatrix} = \begin{pmatrix} A_B^{-1} b \\ 0 \end{pmatrix},$$

$$p^T = \begin{pmatrix} c_B^T & A_B^{-T} c_{N^*}^T \end{pmatrix}$$

Until we reach optimality
i.e., until $\bar{c} \geq 0$

$$p^T = c_B^T A_B^{-1} - \text{dual infeasible.}$$

Complementary Slackness

Thm: Let (x, p) be a primal-feasible,
dual feasible pair.

Then, (x, p) is P-D optimal iff

$$u_i = p_i (a_i^T x - b_i) = 0 \quad \forall i$$

$$v_j = (c_j - p^T A_j) x_j = 0 \quad \forall j$$

pf: $u_i, v_j \geq 0 \quad \forall i, j$.

$$\begin{aligned} \sum_i u_i + \sum_j v_j &= \sum_i p_i a_i^T x - p^T b + c^T x - \sum_j p^T A_j x_j \\ &= \cancel{p^T A x} - p^T b + c^T x - \cancel{p^T A x} \\ &= c^T x - p^T b. \end{aligned}$$

Thus: $u_i = v_j = 0 \quad \forall i, j \quad \underline{\text{iff}} \quad x, p$ are in fact optimal.